

May 28, 2026 12:34 PM GMT

Global Unichip Corp | Asia Pacific

Samsung foundry design service competition creates market share concern; downgrade to EW

WHAT'S CHANGED

Global Unichip Corp (3443.TW)	From	To
Rating	Overweight	Equal-weight

Despite robust CPU demand over the next few years, competition has risen from other foundry peers. Downgrade to EW after the share price rally.

1) Tesla AI chip market share gets diluted by Samsung foundry: GUC stock fell 10% today (vs. Taiex -2%). Investors are concerned about potential competition/ share loss for GUC's existing projects/customers. Given leading-edge capacity shortages and higher TSMC wafer costs, our recent industry checks show that the AI5 packaging configuration could shift from two AI5 chips to one AI5 plus a co-packaged backup AI4 built on Samsung 5nm. What's more, both Samsung and Intel foundries are likely to produce next-gen 2nm AI6 as we noted in our previous [Foundry Floorplan](#). Although we have not yet baked any AI6 revenue into our current GUC forecast, we think it might disappoint some bullish investors who expect GUC to handle the majority of the next-gen Tesla AI project.

2) Google CPU projects could be shared by Samsung foundry as well: Google may also seek other foundry suppliers or potentially other ASIC partners for more capacity. According to our industry checks, Samsung's foundry is also likely to produce next-gen Google CPU (Axion 3) from its SF2 (or 2nm) process, with yields gradually improving. We see potential market share loss for GUC's major CPU business in the coming years, offsetting Google CPU demand strength at GUC. We believe we have baked some conservatism into contribution from GUC's major projects — including Google CPU (2mn units in 2026 and 2.5mn in 2027 and Tesla AI5 (800k units in 2027 and 1.5mn in 2028 — (without AI6). Therefore, we make no changes to our earnings forecasts. We believe potential competition should have limited near-term impact. However, we don't see upside to our current earnings forecasts or price target of NT\$4,888 either.

Move GUC to EW to take profits until we see new project wins: With the medium-/long-term outlook affected and after the share price rally over the past two months (+135% vs. Taiex +33%), we think it's time to take profits, with P/E now at 50x our 2027 EPS. We could turn constructive again if 1) TSMC can allocate more wafers to GUC with price discounts to regain market share, and/or 2) GUC wins new major CSP projects.

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Global Unichip Corp (3443.TW, 3443 TT)				
Greater China Technology Semiconductors Taiwan				
Stock Rating	Equal-weight			
Industry View	Attractive			
Price target	NT\$4,888.00			
Up/downside to price target (%)	6			
Shr price, close (May 28, 2026)	NT\$4,590.00			
52-Week Range	NT\$5,950.00-1,030.00			
Mkt cap, curr (mn)	NT\$619,062			
Avg daily trading value (mn)	NT\$6,101			
Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT\$)**	27.95	58.13	100.85	134.39
Prior EPS (NT\$)**	-	-	-	-
EPS (NT\$)§	27.83	46.05	87.27	115.82
Revenue, net (NT\$ mn)	34,141	80,419	128,426	173,534
P/E	76.0	79.0	45.5	34.2

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

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Still strong chip demand from CPU and AI business

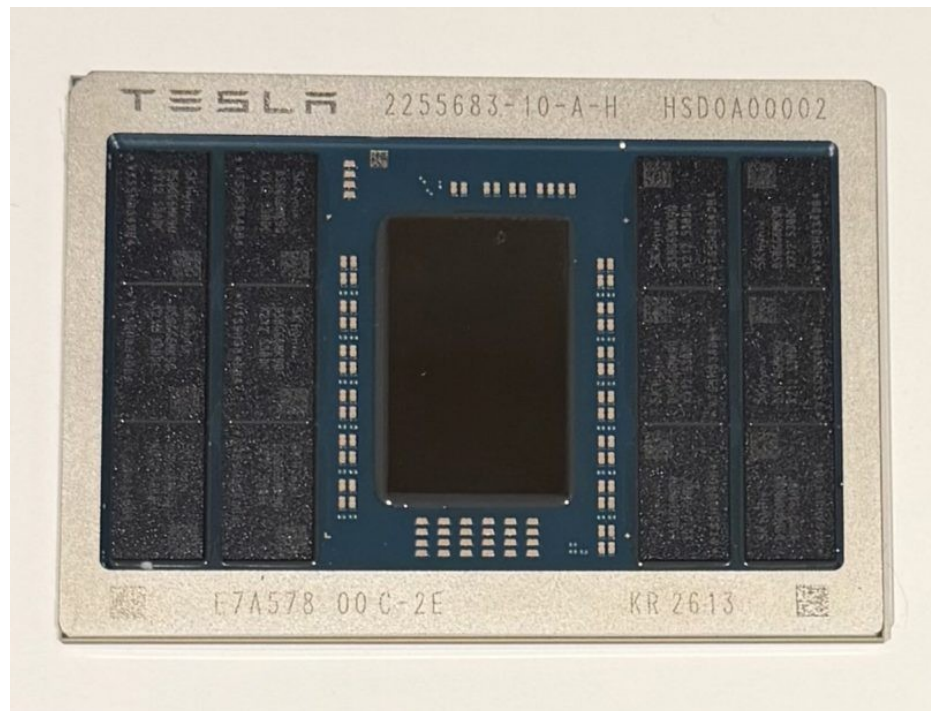
How GUC could participate in TSMC's automotive chip foundry service

In [TSMC: Thoughts about auto semi foundry dynamics; OW \(1 Aug 2025\)](#), we noted that the new AI5 will be manufactured using TSMC's 3nm process node and is expected to deliver 4-5x higher performance than its predecessor, AI4. GUC could help in the design by working closely with TSMC. We believe the Tesla AI project will only get bigger when the new generation chip is introduced. For the next-generation Tesla chip, despite the competition from other foundries, we still expect GUC to get involved in TSMC's version of the AI chip too.

In the same report, we also noted that TSMC is the major autonomous driving chip foundry for merchant chip suppliers such as Horizon Robotics and Nvidia (e.g., the [Thor chip](#)). We believe some of those merchant chips may go through a design service to ensure time to market.

Some Chinese EV OEMs with custom ASIC chip designs are also potential customers for Asian ASIC service providers and TSMC. For example, Alchip (OW) is starting the second 5nm ASIC for Li Auto at TSMC. XPeng is also seeking design service for its new generation ASIC.

Exhibit 1: Tesla AI5



Source: Tesla

Strong Arm-based CPU revenue to dilute gross margin, but still be positive for GUC's EPS

CPU turnkey revenue carries a low gross margin – likely in the single digits – because this is a “Turnkey 2 or 3” project, where GUC provides advanced packaging design support (e.g., silicon interposers and ABF substrates) but does not own the photomasks for the ASIC designs. Our checks suggest that Google CPU gross margin could also be 2-3ppts better than our previous anticipation of 5% given the better order scale at TSMC. We now expect strong Google CPU demand and already saw Axion CPU demand at least 7mn units from 2026-2028, generating close to US\$5bn in revenue (ASP US\$700) for GUC over the same period.

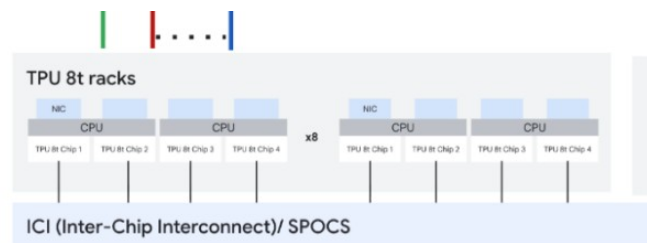
GUC also helps with the Cobalt 200 CPU for Microsoft, an Arm-based design tailored for Microsoft Cloud. Our checks also indicate strong demand for Cobalt at >1mn units in 2027 and onwards. We therefore forecast US\$500-600mn revenue contribution from this in 2027-2028.

Exhibit 2: Both Google 8t and 8i are adopting Axion Arm-based CPU

Feature	TPU 8t	TPU 8i
Primary Workload	Large-scale pre-training	Sampling, serving, and reasoning
Network Topology	3D torus	Boardfly
Specialized Chip Features	SparseCore (Embeddings) & LLM Decoder Engine	CAE (Collectives Acceleration Engine)
HBM Capacity	216 GB	288 GB
On-Chip SRAM (Vmem)	128 MB	384 MB
Peak FP4 PFLOPs	12.6	10.1
HBM Bandwidth	6,528 GB/s	8,601 GB/s (~1.3x of TPU 8t)
CPU Header	Arm Axion	Arm Axion

Source: Google

Exhibit 3: TPU/CPU ratio within TPU 8t rack could be 2:1



Source: Google

Exhibit 4:

We expect US CSP projects to contribute >50% of GUC's 2026e revenue

Revenue (US\$ mn)	2025e	2026e	2027e	2028e	2025 Y/Y	2026 Y/Y	2027 Y/Y	2028 Y/Y
NRE								
Normal NRE projects	76	170	200	220	-49%	124%	18%	10%
Ship-outs	185	108	110	120	68%	-42%	2%	9%
Total NRE	261	278	310	340	0%	7%	12%	10%
Turnkey								
US CSPs	210	1,605	2,700	4,000	600%	664%	68%	48%
Others	621	653	1,050	1,152	31%	5%	61%	10%
Total turnkey	831	2,258	3,750	5,152	65%	172%	66%	37%
Others								
IP	8	17	17	17	-43%	113%	0%	0%
Total revenue	1,100	2,553	4,077	5,509	41%	132%	60%	35%

Source: Company data, e = Morgan Stanley Research estimates

Exhibit 5: GUC: Major Customer Breakdown

Revenue (US\$ mn)	2025e	2026e	2027e	2028e	2025 Y/Y	2026 Y/Y	2027 Y/Y	2028 Y/Y
NRE								
Normal NRE projects	76	170	200	220	-49%	124%	18%	10%
Ship-outs	185	108	110	120	68%	-42%	2%	9%
Total NRE	261	278	310	340	0%	7%	12%	10%
Turnkey								
Microsoft	60	80	150	400	100%	33%	88%	167%
Google	150	1,400	1,750	2,100	NA	833%	25%	20%
Korean AI Start-up	0	20	80	80	NA	NA	300%	0%
xAI/Tesla	0	125	800	1,500	NA	NA	540%	88%
Rivos	0	0	0	0	NA	NA	NA	NA
Amazon Lab 126	0	50	100	100	NA	NA	100%	0%
SK Hynix	50	10	10	10	-38%	-80%	0%	0%
Sandisk	0	20	250	250	NA	NA	1150%	0%
China ADAS	0	20	100	200	NA	NA	400%	100%
Suzhou Centec	54	27	14	7	-50%	-50%	-50%	-50%
Macronix	56	68	81	98	30%	20%	20%	20%
DJI	35	10	5	5	-31%	-71%	-50%	0%
Aspeed	105	121	139	150	50%	15%	15%	8%
Sony	29	26	23	21	-10%	-10%	-10%	-10%
Other turnkey	292	281	248	232	221%	-3%	-12%	-7%
Total turnkey	831	2,258	3,750	5,152	65%	172%	66%	37%
Ex-US CSP turnkey	621	653	1,050	1,152	31%	5%	61%	10%
IP								
Google (GLink)	2	5	5	5	NA	150%	0%	0%
Microsoft (HBM3E)	4	5	5	5	0%	25%	0%	0%
Others	2	7	7	7	-80%	250%	0%	0%
Total revenue	1,100	2,553	4,077	5,509	41.2%	132.1%	59.7%	35.1%

Source: Company data, Morgan Stanley Research estimates

Other major projects in 2026/2027

GUC has indicated that it will have three AI-related projects with US cloud service providers (CSPs) active in 2026. We believe these CSP projects could account for over 50% of 2026e revenue and look for crypto-related momentum to continue at least through 1H26, as the company has already secured foundry capacity support. In our view, this underscores the importance of strong foundry relationships in obtaining scarce leading-edge capacity. Including GUC's other AI start-up engagements and IP licensing, we estimate AI-related revenue will exceed 50% of total revenue in both 2026 and 2027.

- **Microsoft:** In our [previous report](#), we indicated that GUC supports a Microsoft 3nm AI accelerator (Maia 200) TK2. Our industry checks suggest that Maia 200 tape-out should come in May 2026, with mass production later next year. GUC also helps on the Cobalt 200 CPU, Arm-based design tailored for Microsoft Cloud.
- **Amazon AI inference chips:** GUC is helping AWS build the inference chip for AI speakers, which could also start revenue contribution in 2026 and reach around US\$100mn in 2027.
- **SanDisk and Kioxia SSD controllers:** From our industry checks, GUC has also been working with SanDisk on SSD controllers, which should start to contribute some NRE revenue in 2026, and then ~US\$250mn in turnkey revenue in 2027.
- **Chinese ADAS chipmaker:** In addition, GUC's auto customers are not limited to Tesla only. A Chinese ADAS chipmaker could also contribute revenue by 2026 and ramp in 2027.
- **GLink IP licensing:** In 2026, we also expect GUC to keep licensing its die-to-die GLink IP to Google's CPU project, which we estimate could generate US\$3-5mn in annualized revenue.

GUC: Quarterly financials

Exhibit 6: GUC: Quarterly financials

(NT\$ mn)	1Q25	2Q25	3Q25	4Q25	1Q26e	2Q26e	3Q26e	4Q26e	2024	2025	2026e	2027e	2028e
Total revenues	7,024	6,105	8,613	12,400	11,448	14,375	20,638	33,959	25,044	34,141	80,419	128,426	173,534
Q/Q Change	16.6%	-13.1%	41.1%	44.0%	-7.7%	25.6%	43.6%	64.5%					
Y/Y Change	23.4%	-9.2%	30.3%	105.9%	63.0%	135.5%	139.6%	173.9%	-4.6%	36.3%	135.6%	59.7%	35.1%
Cost of Sales	5,051	4,071	6,527	10,038	8,999	11,555	17,014	28,721	16,937	25,687	66,289	106,862	146,040
Percent of Revenues	71.9%	66.7%	75.8%	81.0%	78.6%	80.4%	82.4%	84.6%	67.6%	75.2%	82.4%	83.2%	84.2%
Gross Profit	1,973	2,034	2,086	2,361	2,449	2,820	3,624	5,238	8,108	8,454	14,131	21,564	27,493
Gross Margin	28.1%	33.3%	24.2%	19.0%	21.4%	19.6%	17.6%	15.4%	32.4%	24.8%	17.6%	16.8%	15.8%
Total Opex	878	1,090	1,031	1,105	1,183	1,237	1,344	1,525	4,305	4,104	5,289	5,793	6,402
Percent of Revenues	12.5%	17.9%	12.0%	8.9%	10.3%	8.6%	6.5%	4.5%	17.2%	12.0%	6.6%	4.5%	3.7%
R&D	800	837	813	880	864	884	919	980	3,223	3,330	3,647	3,751	3,985
Percent of Revenues	11.4%	13.7%	9.4%	7.1%	7.5%	6.1%	4.5%	2.9%	12.9%	9.8%	4.5%	2.9%	2.3%
General & Adm Exp.	-18	159	125	129	170	175	190	200	674	395	735	740	855
Percent of Revenues	-0.3%	2.6%	1.5%	1.0%	1.5%	1.2%	0.9%	0.6%	2.7%	1.2%	0.9%	0.6%	0.5%
Selling Expenses	97	94	93	96	149	178	235	345	407	379	907	1,303	1,562
Percent of Revenues	1.4%	1.5%	1.1%	0.8%	1.3%	1.2%	1.1%	1.0%	1.6%	1.1%	1.1%	1.0%	0.9%
Operating Income	1,095	944	1,055	1,257	1,266	1,583	2,280	3,713	3,803	4,350	8,842	15,770	21,091
Operating Margin	15.6%	15.5%	12.2%	10.1%	11.1%	11.0%	11.0%	10.9%	15.2%	12.7%	11.0%	12.3%	12.2%
Total Non-operating Income (loss)	48	-15	-40	99	95	95	95	95	260	93	382	232	232
Profit Before Taxes	1,143	929	1,015	1,356	1,362	1,678	2,375	3,809	4,062	4,443	9,224	16,002	21,323
Percent of Revenues	16.3%	15.2%	11.8%	10.9%	11.9%	11.7%	11.5%	11.2%	16.2%	13.0%	11.5%	12.5%	12.3%
Taxes	181	147	148	196	204	252	356	571	612	673	1,384	2,400	3,198
Tax Rate	15.9%	15.9%	14.6%	14.5%	15.0%	15.0%	15.0%	15.0%	15.1%	15.2%	15.0%	15.0%	15.0%
Minor interest	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Net Income to Parent	961	782	867	1,160	1,158	1,426	2,019	3,237	3,451	3,770	7,840	13,602	18,125
Percent of Revenues	13.7%	12.8%	10.1%	9.4%	10.1%	9.9%	9.8%	9.5%	13.8%	11.0%	9.7%	10.6%	10.4%
Reported Basic EPS (NT\$, TW GAAP)	7.13	5.80	6.43	8.60	8.58	10.58	14.97	24.00	25.63	27.95	58.13	100.85	134.39
Y/Y Change	45%	-14%	-16%	37%	20%	82%	133%	179%	-2%	9%	108%	73%	33%

Source: Company data, Morgan Stanley Research (e) estimates

Price target remains unchanged at NT\$4,888

Our price target remains unchanged. All of our key valuation assumptions are also unchanged, including cost of equity of 9.2% (a beta of 1.2, a risk premium of 6%, and a risk-free rate of 2%), intermediate growth rate of 17%, and terminal growth rate of 5.0%.

Our bull and bear case values are also at NT\$6,645 and NT\$3,010, respectively.

Exhibit 7: GUC: Residual income model

NT\$million	2026e	2027e	2028e	2029e	2030e	2031e	2032e	2033e	2034e	2035e	2036e	2037e
Total Equity	18,707	28,053	38,794	47,795	58,326	70,647	85,063	101,930	121,664	144,752	171,766	203,372
Net Profit	7,840	13,602	18,125	21,206	24,811	29,029	33,964	39,737	46,493	54,397	63,644	74,464
ROAE	49.6%	58.2%	54.2%	49.0%	46.8%	45.0%	43.6%	42.5%	41.6%	40.8%	40.2%	39.7%
Residual Income	5,216	9,162	12,631	15,433	17,952	20,890	24,320	28,327	33,012	38,489	44,895	52,387
Spread	40.4%	49.0%	45.0%	39.8%	37.6%	35.8%	34.4%	33.3%	32.4%	31.6%	31.0%	30.5%
Ending Equity Capital	18,707											
PV of Forecast Period	138,930											
PV of Continuing Value	497,414											
Equity Value	655,050											
No. of Shares	134											
Projected Price	4,888											

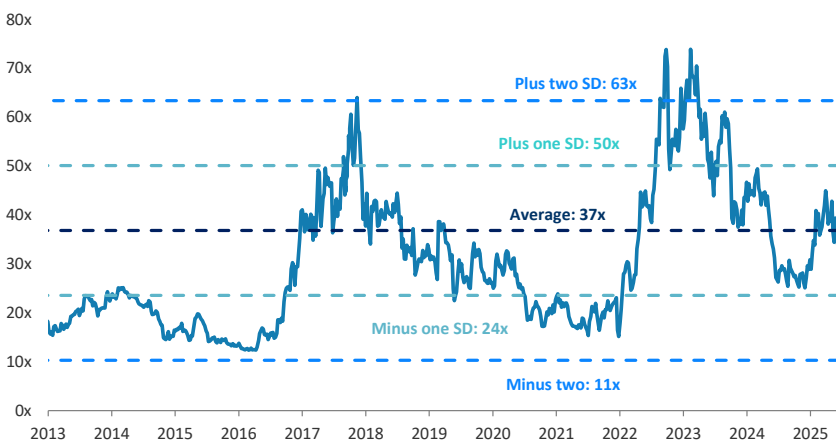
Source: Company data, Morgan Stanley Research (e) estimates

Relative valuation

GUC's share price has risen nearly 135% since April 2025 given the strong demand from Google CPU and Tesla AI5 (the TAIEX is up 33% over the same period).

Our price target of NT\$4,888 implies 48x our 2027 EPS estimate, lower than +1 SD from GUC's historical NTM P/E since 2013. We maintain our view that the design service market will proliferate with AI ASIC outgrowing AI GPU in the long term. However, we think it's time to take profits, with the stock now trading at 50x our 2027 EPS estimate with potential competition.

Exhibit 8: GUC: NTM rolling P/E



Source: Company data, FactSet, Morgan Stanley Research estimates

Risk Reward – Global Unichip Corp (3443.TW)

Facing medium-/long-term competition for major projects

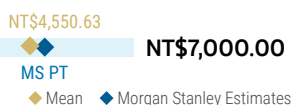
PRICE TARGET NT\$4,888.00

Base case, residual income model. Key assumptions: a cost of equity of 9.2%, an intermediate growth rate of 17%, and a terminal growth rate of 5%.

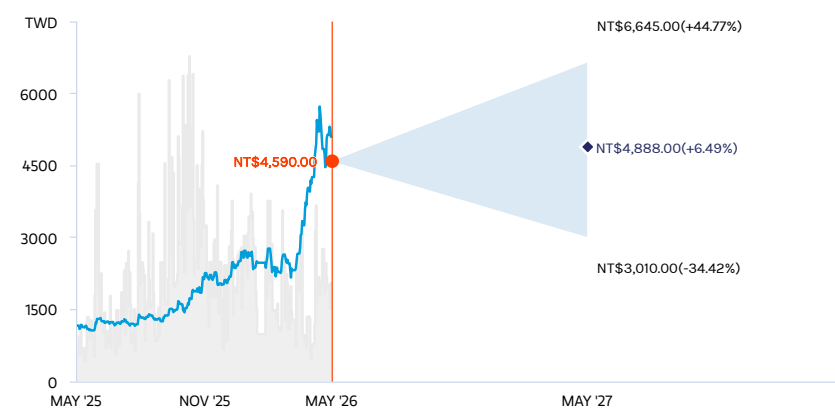
Consensus Price Target Distribution

NT\$2,500.00 NT\$4,550.63 NT\$7,000.00

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



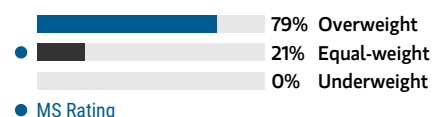
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Besides strong Google CPU revenue contribution in 2026, we see 2027 revenue upside from Tesla AI5, Amazon edge AI chips, and SanDisk SSD controllers.
- We think it's time to take profits, with the stock now trading at 50x our 2027 EPS estimate and the company facing potential competition.
- We maintain our view that the design service market will proliferate with AI ASIC outgrowing AI GPU in the long term.
- Despite the slower recovery of non-AI business, we still appreciate GUC's growing exposure to cloud AI semis (including AI accelerators, AI server CPU, and BMC chips).

Consensus Rating Distribution



● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Pricing Power: *Negative*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

NT\$6,645.00

66x 2027e EPS

Multiple leading-edge NRE projects successfully translate into volume production with healthy margin profiles. Many AI chipsets or other projects start to proliferate with substantial end demand, with customers' interest in migrating to more advanced nodes. Networking switch demand is stronger than expected, while SSD controller project contributions are sustainable into 2025. 3/5nm turnkey revenue growth is stronger than expected in 2026 and beyond.

BASE CASE

NT\$4,888.00

48x 2027e EPS

Networking switch turnkey projects volume remains stable. A few new leading-edge turnkey projects start to show decent shipments. TSMC's leading-edge tape-out numbers increase steadily, while some "turnkey 2/3" (TK2 and TK3) projects contribute to NRE and turnkey revenue. CSP (AI) revenue contribution should be the major growth driver for 2026 and 2027, as well as the auto chip and SSD controller projects.

BEAR CASE

NT\$3,010.00

30x 2027e EPS

AI chipsets and other projects lack concrete end demand, resulting in delayed tape-outs. TSMC's leading-edge tape-out numbers decrease or some customers turn directly to TSMC. BMC and SSD project losses arise from customers' in-sourcing. Fiercer-than-expected talent competition leads to higher opex to retain R&D engineers. High volume turnkey projects carry lower-than-expected gross margin.

Risk Reward – Global Unichip Corp (3443.TW)

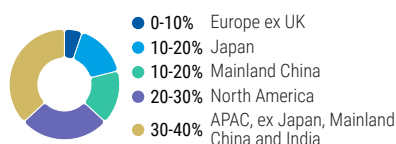
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
NRE revenue Y/Y (%)	(4.8)	5.4	11.7	9.7
Turnkey revenue Y/Y (%)	59.2	178.6	65.6	37.2
Turnkey revenue (NT\$, mn)	25,736	71,696	118,714	162,907
Turnkey revenue contribution (%)	75.4	89.2	92.4	93.9

INVESTMENT DRIVERS

- ASIC/ASSP industry demand
- Technology migration to more advanced nodes
- New projects driven by China's semi localization efforts

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

3/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- 5nm, 3nm and advanced packaging customers decide to accelerate tape-outs.
- AI and other high-growth projects gain substantial customer traction.
- 5nm turnkey momentum is stronger than expected.

RISKS TO DOWNSIDE

- AI and other high-growth projects lose significant customer traction.
- Customers' inventory correction is longer than expected.
- 5nm and below ASIC projects contribution is lower than expected.

OWNERSHIP POSITIONING

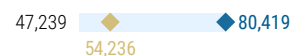
Inst. Owners, % Active **57.8%**

Source: Refinitiv, Morgan Stanley Research

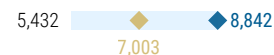
MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

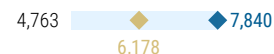
Sales / Revenue
(NT\$, mn)



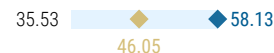
EBIT
(NT\$, mn)



Net income
(NT\$, mn)



EPS
(NT\$)



◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Financial summary

Income Statement

NT\$m (Years End Dec)	2024	2025	2026e	2027e	2028e
Net sales	25,044	34,141	80,419	128,426	173,534
COGS	(16,937)	(25,687)	(66,289)	(106,862)	(146,040)
Gross profit	8,108	8,454	14,131	21,564	27,493
Operating expenses	(4,305)	(4,104)	(5,289)	(5,793)	(6,402)
Operating income	3,803	4,350	8,842	15,770	21,091
Non-operating income	260	93	382	232	232
Pre-tax income	4,062	4,443	9,224	16,002	21,323
Income tax	612	673	1,384	2,400	3,198
Minority Interest	0	0	0	0	0
Reported net Income	3,451	3,770	7,840	13,602	18,125
Adj.wtd.avg.shrs(m)	134	134	134	134	134
Reported EPS (NT\$)	25.63	27.95	58.13	100.85	134.39
Modelware EPS (NT\$)	25.63	27.95	58.13	100.85	134.39

Balance Sheet

NT\$m (Years End Dec)	2024	2025	2026e	2027e	2028e
Cash	10,427	6,650	4,944	7,810	15,266
Mkt Securities	2,980	2,082	2,082	2,082	2,082
AR/NR	2,007	3,417	6,389	10,203	13,787
Inventory	2,794	10,148	18,056	25,317	29,454
Other	6,271	3,440	3,440	3,440	3,440
Current Assets	24,480	25,737	34,912	48,852	64,030
Long-term investments	0	0	0	0	0
Fixed assets	942	1,347	1,347	1,347	1,347
Deferred assets	37	24	24	24	24
Other assets	850	1,335	1,335	1,335	1,335
Total Assets	26,309	28,443	37,617	51,557	66,735
S/T borrowings	0	0	0	0	0
AP/NP	1,691	4,126	7,506	12,101	16,537
Other ST liabilities	12,977	10,907	10,907	10,907	10,907
LT debt	73	169	169	169	169
Other LT liabilities	273	328	328	328	328
Common shares	1,340	1,340	1,340	1,340	1,340
Total Liabilities	15,015	15,530	18,910	23,505	27,941
Additional capital	1,812	2,158	2,158	2,158	2,158
Retained earning	8,145	9,431	15,225	24,570	35,312
Other shareholders' equity	(3)	(16)	(16)	(16)	(16)
Total Equity	11,294	12,913	18,707	28,053	38,794
Total Liab. & Shrhldr's Equity	26,309	28,443	37,617	51,557	66,735

E = Morgan Stanley Research Estimates

Source: Morgan Stanley Research, Company Data

Cash Flow Statement

NT\$m (Years End Dec)	2024	2025	2026e	2027e	2028e
Cashflow from Operations	6,242	(1,143)	613	7,395	15,113
Net profits	3,451	3,770	7,840	13,602	18,125
Depreciation	247	273	273	273	273
Working Capital Change	1,925	(5,930)	(7,500)	(6,480)	(3,285)
Other adjustments	619	744	0	0	0
Cashflow from Investing	(1,522)	(398)	(273)	(273)	(273)
Capex	(471)	(840)	(273)	(273)	(273)
Change of LT Investment	0	0	0	0	0
Change of ST Investment	(850)	688	0	0	0
Other adjustments	(201)	(246)	0	0	0
Cashflow from financing	(1,962)	(2,227)	(2,046)	(4,256)	(7,383)
Increase in L/T debt	(40)	96	0	0	0
Increase in S/T debt	0	0	0	0	0
Cash Dividend Paid	(1,876)	(2,144)	(2,046)	(4,256)	(7,383)
Dir& Emp Bonus Paid	0	0	0	0	0
Issuance of stock	0	0	0	0	0
Other adjustments	(46)	(178)	0	0	0
Exchange rate adjustment	31	(10)	0	0	0
Net change in cash	2,790	-3,778	-1,706	2,866	7,457

Financial Ratios

	2024	2025	2026e	2027e	2028e
Growth(%)					
Turnover	-4.6	36.3	135.6	59.7	35.1
Operating profits	-4.2	14.4	103.3	78.4	33.7
Pretax profits	-2.2	9.4	107.6	73.5	33.3
Net profits	-15.8	9.2	108.0	73.5	33.3
EPS	-1.5	9.1	108.0	73.5	33.3
Margins (%)					
Gross Margin	32.4	24.8	17.6	16.8	15.8
Operating Margin	15.2	12.7	11.0	12.3	12.2
Pretax Margin	16.2	13.0	11.5	12.5	12.3
Net Profit	13.8	11.0	9.7	10.6	10.4
Return (%)					
ROAE	32.9	31.1	49.6	58.2	54.2
ROAA	14.6	13.8	23.7	30.5	30.6
Gearing (%)					
Net Debt/Equity	(92.3)	(51.5)	(26.4)	(27.8)	(39.4)
Liabilities/Equity	132.9	120.3	101.1	83.8	72.0
Ratios (X)					
Current ratio	1.7	1.7	1.9	2.1	2.3
Quick ratio	0.8	0.7	0.6	0.8	1.1
Others					
AR/NR Turnover (days)	29	29	29	29	29
Inventory Turnover (days)	82	92	82	72	62
AP Turnover (days)	36	41	41	41	41
Cash Conversion (days)	75	80	70	60	50

Valuation Methodology and Risks

TSMC (2330.TW)

Base case, residual income model. Key assumptions: a cost of equity of 9.2% (beta of 1.2, risk-free rate of 2.0% and risk premium of 6.0%), an intermediate growth rate of 10.5%, and a terminal growth rate of 4.0%.

Risks to Upside

- TSMC charges large customers more to keep its gross margin above 56% in the long term.
- AI semi demand grows more significantly than expected, while TSMC maintains high market share in leading-edge foundry business.
- Outsourcing from Intel CPU increases in 2025-27.

Risks to Downside

- Inventory correction occurs in 2026.
- Demand for leading edge technologies weakens.
- Costs of overseas fabs grow significantly.

Alchip Technologies Ltd (3661.TW)

Base case value, derived from a residual income model. We assume the cost of equity is constant at 10.4% (2% risk-free rate, 6% risk premium, 1.4 beta). Our intermediate growth rate and terminal growth rates are 16% and 5%, respectively.

Risks to Upside

- Demand for Chinese GPUs and AI chipsets takes off earlier than expected.
- NRE demand accelerates.
- Alchip wins more US hyperscaler HPC projects.

Risks to Downside

- China's HPC localization progress slows down.
- AI chipset progress is much slower to develop.
- NRE demand decelerates.
- Pricing competition intensifies.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
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(as of April 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1546	42%	467	51%	30%	709	44%
Equal-weight/Hold	1568	43%	358	39%	23%	715	44%
Not-Rated/Hold	4	0%	0	0%	0%	1	0%
Underweight/Sell	555	15%	84	9%	15%	202	12%
Total	3,673		909			1627	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Stock Price, Price Target and Rating History (See Rating Definitions)

Alchip Technologies Ltd (3661.TW) - As of 05/28/26 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 5/1/21 : E/I; 5/14/21 : O/I; 10/12/21 : O/C; 10/4/22 : O/A; 2/22/23 : O/I; 7/7/23 : O/A; 7/21/24 : O/I; 7/14/25 : O/A

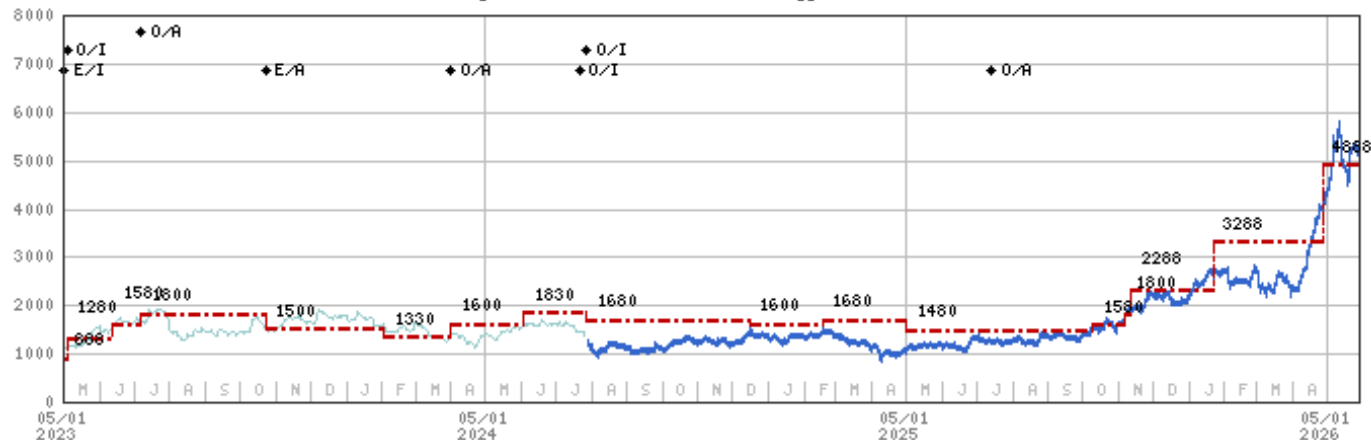
Price Target History: 4/12/21 : 780; 9/23/21 : 980; 10/8/21 : 1090; 10/30/21 : 1190; 2/10/22 : 1280; 3/7/22 : 1420; 10/18/22 : 1280; 3/13/23 : 1340; 4/26/23 : 1380; 5/10/23 : 1480; 6/12/23 : 1800; 7/3/23 : 2130; 7/7/23 : 2280; 8/14/23 : 2330; 9/12/23 : 2880; 11/3/23 : 3180; 1/2/24 : 3880; 2/14/24 : 4880; 6/10/24 : 4280; 9/26/24 : 3800; 12/16/24 : 3988; 7/3/25 : 4088; 8/12/25 : 4588; 9/17/25 : 4288; 11/5/25 : 4388; 5/7/26 : 5088

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target --- No Price Target Assigned (NA)
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Global Unichip Corp (3443.TW) - As of 05/28/26 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 5/1/21 : U/I; 10/12/21 : U/C; 10/4/22 : U/A; 10/28/22 : O/A; 12/28/22 : E/A; 2/22/23 : E/I; 5/4/23 : O/I; 7/7/23 : O/A; 10/23/23 : E/A; 4/1/24 : O/A; 7/21/24 : O/I; 7/27/24 : O/I; 7/14/25 : O/A

Price Target History: 1/12/21 : 300; 5/20/21 : 290; 10/29/21 : 450; 10/28/22 : 550; 12/28/22 : 700; 4/6/23 : 888; 5/4/23 : 1280; 6/12/23 : 1580; 7/7/23 : 1800; 10/23/23 : 1500; 2/3/24 : 1330; 4/1/24 : 1600; 6/3/24 : 1830; 7/27/24 : 1680; 12/16/24 : 1600; 2/17/25 : 1680; 5/1/25 : 1480; 10/9/25 : 1580; 11/7/25 : 1800; 11/12/25 : 2288; 1/22/26 : 3288; 4/27/26 : 4888

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target — No Price Target Assigned (NA)

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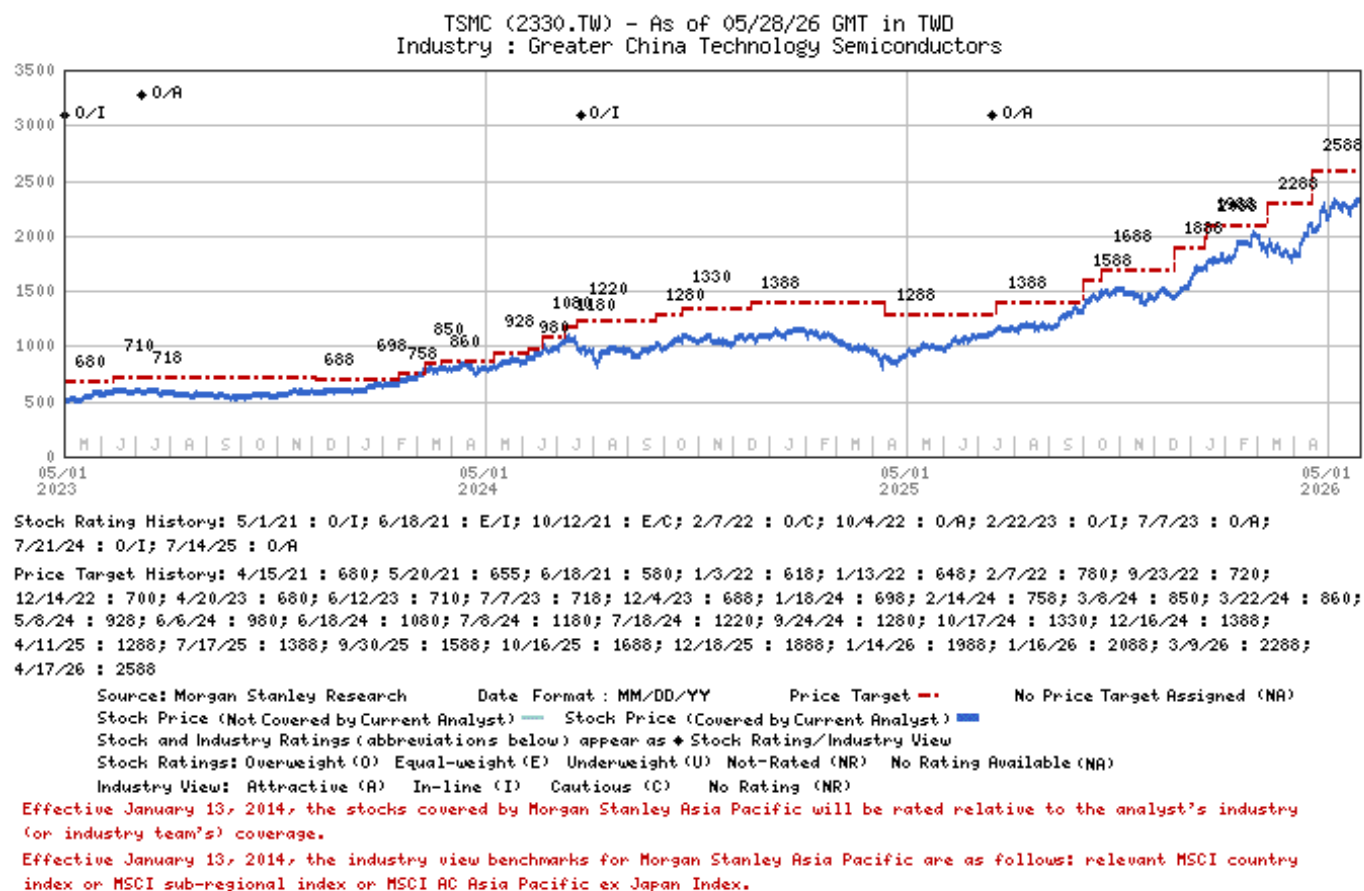
Stock and Industry Ratings (abbreviations below) appear as ♦ Stock Rating/Industry View

Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)

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INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/28/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US\$88.64
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb453.12
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT\$165.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT\$4,300.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT\$627.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,391.50
Global Unichip Corp (3443.TW)	E (05/28/2026)	NT\$4,590.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT\$927.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT\$535.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK\$170.00
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK\$546.50
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT\$320.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb117.09
MediaTek (2454.TW)	O (11/28/2025)	NT\$4,410.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb699.70
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT\$324.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb659.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb101.90
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT\$2,450.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb121.50
Silergy Corp. (6415.TW)	U (05/19/2026)	NT\$620.00
SMIC (0981.HK)	O (10/21/2025)	HK\$88.25
TSMC (2330.TW)	O (02/07/2022)	NT\$2,295.00
UMC (2303.TW)	O (05/19/2026)	NT\$142.00
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT\$161.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT\$528.00

Daisy Dai, CFA

ASMP Ltd (0522.HK)	O (07/24/2025)	HK\$199.70
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb69.80
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT\$161.00
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb128.58
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb38.44
Innoscence (2577.HK)	E (10/13/2025)	HK\$75.55
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb84.74
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK\$35.50
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb163.55
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb143.00
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb86.30
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb44.00
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb114.30

Daniel Yen, CFA

AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT\$1,095.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT\$1,440.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT\$17,835.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT\$125.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb179.13
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb483.00
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT\$160.00
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK\$458.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb260.98
Novatek (3034.TW)	U (02/04/2026)	NT\$478.50
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT\$188.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT\$845.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT\$80.70
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT\$594.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb66.09
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT\$144.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT\$121.50
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT\$289.50

Duan Liu

Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb147.80
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb548.01

Tiffany Yeh

AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT\$1,240.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT\$815.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US\$20.65
Hon Precision (7769.TW)	O (04/17/2026)	NT\$7,700.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT\$5,915.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US\$284.98
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT\$9,120.00

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* Historical prices are not split adjusted.